

In the previous class, you have studied about many changes taking place in matter such as; melting, freezing, boiling, evaporation and condensation. Why these changes take place? Can you recall what causes these changes? Were these changes reversible or irreversible? Were these changes in the physical properties, appearance and state or in the chemical composition? Have you observed a burning candle? Is this change reversible or irreversible? Can you get the same quantity of wax after the burning of candle? Does any new substance formed during burning? How can we apply the process of changes in matter in our daily life? Let us explore.

In this Chapter you will learn about:

- Physical and chemical changes.
- Applications of chemical changes and processes (changing raw materials into useful products hydrocarbons, fats, fertilizers, plastics)
- Reversible and irreversible Changes.

All the students will be able to:

- ✓ Differences between physical and chemical changes.
- ✓ Identify the physical and chemical changes taking place in environment.
- ✓ Explain the use of hydrocarbons as fuels.
- ✓ Explain the physical and chemical properties of fertilizers, which make them useful in agriculture.
- ✓ Discuss harmful effects of improper use of fertilizers.
- ✓ Describe the processes in which vegetable oil changes into fat.
- ✓ Describe the simple chemical process for manufacture of plastics.
- ✓ Distinguish between reversible and irreversible changes in materials.
- ✓ Identify a variety of reversible and irreversible changes in materials in their surroundings.



Fig 7.1(a) : Physical changes.



Fig 7.1(b) : Chemical changes.

PHYSICAL AND CHEMICAL CHANGES

- ✓ Differentiate between physical and chemical change.
- ✓ Distinguish between reversible and irreversible changes in materials.

Many changes occur in matter around us. Some changes are very rapid and some are very slow. Some changes are reversible and others are not. These changes in matter are of two types:

1. Physical change.
2. Chemical change.

How would you recognize a physical change?

How would you recognize a chemical change?

Table 7.1 The sign of physical and chemical changes.

Signs of a physical change	Signs of a chemical change
• Change that can be reversed. • The substance does not turn into a new substance.	• A release of gas. • Change in colour. • Change in temperature. • Change in smell. • Production of a new material.

(1) Physical Change:

Physical change is a change in physical appearance or state of a substance. It does not affect the composition and the substance remains same before and after the change, for example breaking of bottle and cutting of paper into small pieces, only the shape and size of the glass bottle and paper is changed and no new substance is formed.

DO YOU KNOW?

Physical and chemical changes happen in the world around you and not just in a science laboratory. Matter changes its states through a process called physical change. Matter also interacts with other matter to form new substances through a process called a chemical reaction or chemical change.

INVESTIGATE

Observe your surrounding:
Make a list of physical and chemical changes happening all around you.



Fig 7.2 Breaking of glass bottle



Fig 7.3 Cutting of paper



Fig 7.4: Crumpling of Aluminium foil

All physical changes are related to changes in physical properties such as change in size, shape, texture, volume, weight and temperature.

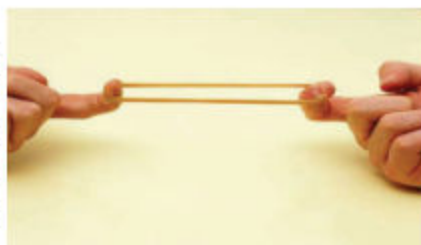


Fig 7.5 Stretching of rubber band

Physical change can be very well explained by the changes occurring in the state of water, such as melting, evaporation, freezing and condensation. (see Fig 7.6).

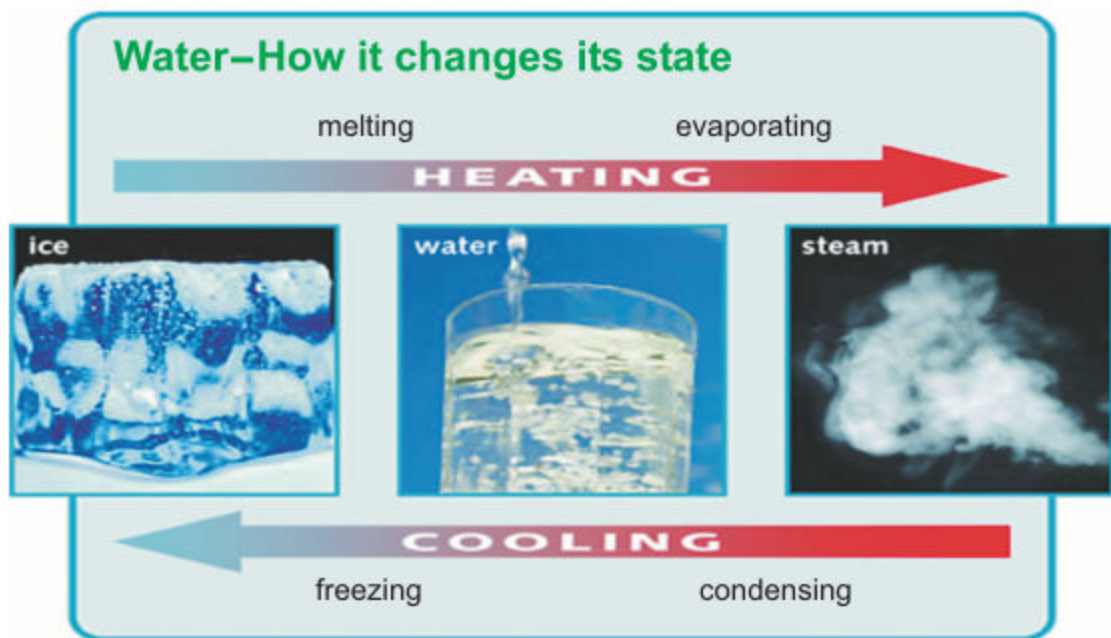


Figure 7.6: Three states of the water

Ice, water and steam are the same chemical substances in different physical states. As steam can be changed back to water on cooling and ice can be changed back to water by heating.

Is dissolving of salt in water a physical or chemical change? Can we obtain salt and water in its original form? When salt is dissolved in water, no new substances are formed. You can easily separate salt (solute) from water (solvent) by evaporation.



Figure 7.7: Dissolving salt in water

It means in a physical change, the composition of matter does not change and material keeps its original composition. The tearing and scrambling a paper, melting of chocolate or ice, boiling, breaking, moulding of clay and metals dissolving sugar and salt in water and slicing fruit, forming of frost from water vapours in the air are examples of physical changes because only appearance, colour, size, volume, shape or form, amount of heat or temperature is changed and composition remains the same. From the above examples it can be concluded that physical changes are temporary, reversible and form no new substance.

(2) Chemical Change:

Matter can undergo chemical changes. In a chemical change a substance is transformed into a different substance with new composition. A chemical change is a process where one or more substances are changed into one or more new and different substances. You have learnt in the previous class that plants make glucose ($C_6H_{12}O_6$) from Carbon dioxide (CO_2), and water (H_2O) in the presence of sunlight. This is chemical change as glucose a new chemical substance is formed by a chemical reaction.

What will happen when you burn a piece of paper?

If you burn the same piece of paper which you have cut, it will form different substances such as ash and carbon dioxide, which cannot change into paper again.

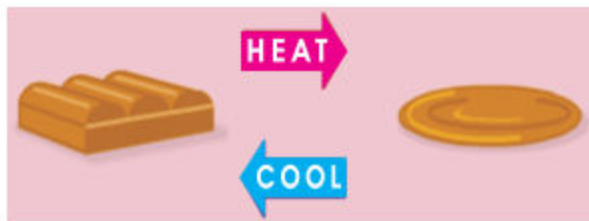


Fig 7.8 Chocolate melting in a liquid by heating and solidifying by cooling

DO YOU KNOW?

Every time you eat something, cook or clean, several chemical changes take place. Chemical changes also take place when you take a breath, take medicine and even light a match.

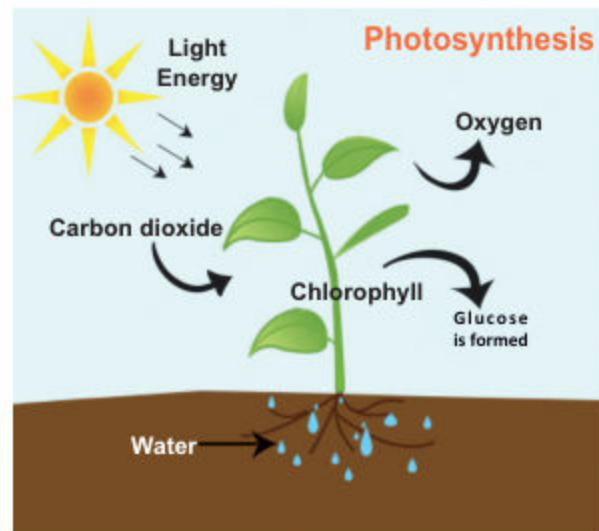


Figure 7.9: Photosynthesis

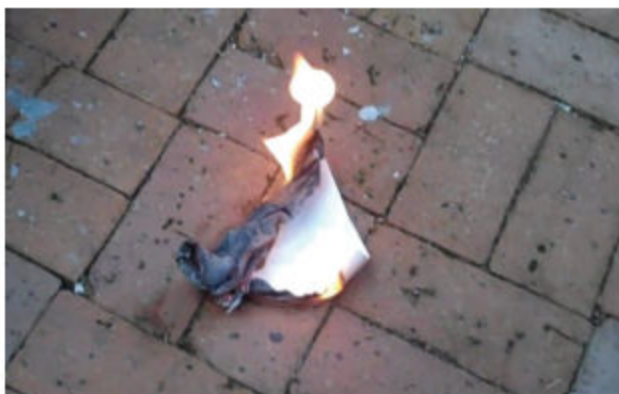


Figure 7.10: Burning of paper

Have you seen a burning candle, how does a burning candle undergo changes in matter and its state? Observe a burning candle, you will see the light, feel the heat and see the melting of the candle wax.

When a candle burns for a while, it eventually gets smaller. Where does the candle wax go? In the process of burning, the heat is melting the wax and changing it into liquid wax. This liquid wax will again convert into solid wax upon cooling. After some time you will smell wax gas. The liquid wax is forming wax gas. Candle wax is composed of chains of connected carbon atoms surrounded by hydrogen atoms.

Upon burning of the candle some wax is also converting into water and carbon dioxide which dissipate in the air. In this way both physical and chemical changes are taking place in the burning candle.



Figure 7.11 Burning candle

DO YOU KNOW?

The most popular type of candle wax used today is paraffin wax. The chemical composition of paraffin wax is commonly referred to as $C_{25}H_{52}$ (Carbon and Hydrogen). However, the actual number of carbon atoms can typically range from 22 to 27. A wax molecule is known as a long hydrocarbon with its general chemical formula being C_nH_{2n+2} , with n being a varying number of carbon atoms (22-27). Although the chemical composition of the wax is always carbon and hydrogen, the actual number of atoms will vary based on the exact origin of the wax.

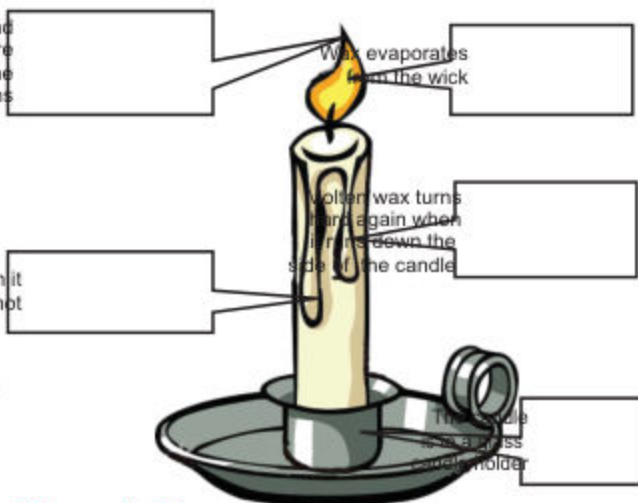


Figure 7.12

Some examples of chemical changes are burning of wood, cooking raw food, bursting of crackers, rusting of iron, digestion of food, respiration in living things and photosynthesis in plants. Chemical changes are permanent and irreversible. In order to understand chemical change, perform a simple activity of rusting of iron.

Activity 7.1: Investigating rusting of iron nail/ wire wool (work in groups of 3-5).

What I need:

- 3 small empty jars with lids/ test tubes with corks.
- Iron nails or wire wool.
- Tap water and boiled water.
- Calcium Chloride

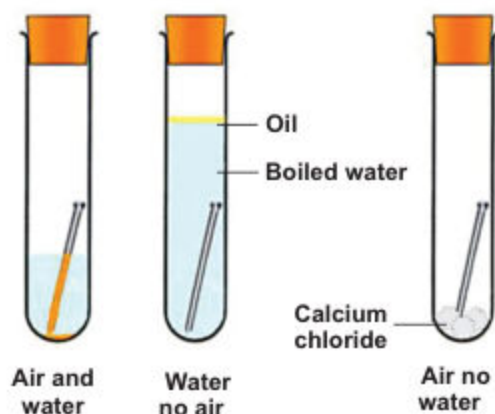


Figure 7.13

What to do:

1. Line up 3 glass jars/ test tubes and mark or label them as 1, 2, and 3.
2. Put one/two nails / a bit of wire wool in each container.
3. Pour some water into container number '1' and put the lid on it.
4. Pour some boiled water into container number '2', add oil and put the lid on it.
5. Add Calcium Chloride in container number '3' and put the lid on.
6. After one day observe containers and write your observations of each container.

What I observed:

Container No.1: _____

Container No.2: _____

Container No.3: _____

Activity Questions:

In which container/test tube the rusting was happening first?
Give reasons.

In which container/test tube the nails did not rust? Give reasons.

Why boiled water and oil is used in container/test tube No.2?

Why calcium chloride was added in container/test tube No.3?

What will happen if calcium chloride is not added?

What I conclude:

Table 7.2: Differences between physical and chemical changes

Physical Changes	Chemical Changes
No new substances are formed. Substance/material remains same and does not lose its major properties or identity.	New substances are formed with different properties than the parent substances.
Generally reversible.	Generally irreversible.
Usually not accompanied by energy change (except in changes in states).	Usually accompanied by energy change.
Examples: changes in states, dissolving sugar in water, crushing a chalk into powdered form and melting of ice-cream.	Examples: burning of fuel, rusting of iron and heating sulphur with iron fillings to produce iron sulphide.

PHYSICAL AND CHEMICAL CHANGES TAKING PLACE IN THE ENVIRONMENT

- ✓ Identify the physical and chemical changes taking place in environment.
- ✓ Identify a variety of reversible and irreversible changes in materials in their surroundings.

Many physical and chemical changes are part of our daily life. After washing the clothes we hang them to dry, all the water from clothes evaporates, which is a physical process. Cooking of breakfast,

lunch and dinner is a chemical process. Dissolving sugar in water is a physical change because no new substance is formed; water and sugar can be separated through a simple process of evaporation.

Did you observe vehicles and factories emitting toxic gasses which are the major cause of acid rain? Look at the colour of Quid-e-Azam's Mausoleum. Why white colour of marble has changed? Can you identify few more examples of physical and chemical changes taking place in the environment? Discuss in your class.

Physical and chemical changes can also be useful. Many of the life processes which take place in the body such as breathing, digestion, movement and respiration involve physical and chemical changes, without them we would not be able to survive.

DO YOU KNOW ?

Explosion of firework is a chemical change. Heat, light, sound, unpleasant gases are produced in such explosions and they can seriously harm you.

A change in which one or more new substances are formed is called a chemical change. A chemical change is also called a chemical reaction.

HYDROCARBON

✓ Explain the use of hydrocarbons as fuel.

Hydrocarbons are compounds made up of hydrogen and carbon, which are one of the types of organic compounds. Organic compounds are the compounds of carbon. Simplest hydrocarbon is methane CH_4 (natural gas) or sui gas which is used in homes and industries as a fuel. Hydrocarbons such as petrol, diesel and furnace oil are used as fuels and their burning is an example of chemical change. During this process energy is produced in the form of heat, and light.

EXPLORATION

Supplies of fossil fuels are limited and are being used at alarming rate. They will not last forever. The fact we are running out of fossil fuels means that we must plan for the future. Suggest some alternatives to overcome this problem? How can we make the best possible use of the energy we get from the fossil fuels?

Activity 7.2: Investigate the melting of wax.

What happens when you melt wax?

Is the change reversible? What type of change is this?

Chemical changes are very important in our lives. All new substances are formed as a result of chemical changes. For example, useful new materials, such as chemical fertilizers, vegetable ghee, plastics and detergents, are produced by chemical reactions. Indeed, every new material is discovered by studying chemical changes.

INVESTIGATE

Heat can bring about chemical reactions and chemical reactions may produce heat. Is this also true for light and electricity?

FERTILIZERS

- ✓ Explain the physical and chemical properties of fertilizers, which make them useful in agriculture.

The substances that are added in the soil to make up any deficiency of nutrients are called fertilizers. Fertilizers are very important for growing crops; they release nutrients into the soil, as they contain required chemicals. However, physical characteristics and nutrient content (chemicals) of fertilizers both are important. They determine how easily and uniformly the fertilizers spread during application. If fertilizers form lumps or dust and accumulate too much water then they are not of good quality.

Following are the physical and chemical properties of fertilizers:

1. Particle size:

Particles of different fertilizers are of different sizes. Fertilizers which have smaller particles dissolve in water faster. Thus, nutrients are quickly released. Particle sizes also affect application and storage of the fertilizer. Hard particles are better than soft particles because they release nutrients gradually.

2. Density:

Density depends on how closely packed the particles are? A low density fertilizer will take up more space than the same weight of a high density one.

3. Granule hardness:

Fertilizer particles need to be hard enough to with stand the pressure of handling and storing them. Granule hardness will depend on the chemical composition of the fertilizer as well as its other physical properties such as the shape of the particles and how much moisture they contain.

4. Moisture content:

Most fertilizers absorb water to a certain extent. However, if they take in too much water, it can be problematic. Moisture absorption depends on the chemical composition of the fertilizer, environmental conditions and the shape and size of the particles. Granules with larger surface area absorb relatively more water.

Chemical Properties of Fertilizers:

Plants have various nutrient needs, depending upon the species, the age, and location of the plant. They need sixteen nutrients for healthy growth. The major nutrients are calcium, magnesium, potassium, carbon, phosphorus, sulphur, hydrogen, nitrogen and oxygen. Addition of these nutrients for the proper growth of crops is called soil maintenance. Mostly, fertilizers are made from carbon, hydrogen and oxygen, few other elements are also involved like: phosphorus, nitrogen, potassium, calcium, sulphur etc.

Table 7.3 shows some common fertilizers.

Name of Fertilizer	Chemical Formula	Name of Fertilizer	Chemical Formula
Ammonium Nitrate	NH_4NO_3	Ammonium Sulphate	$(\text{NH}_4)_2\text{SO}_4$
Ammonium Phosphate	$(\text{NH}_4)_3\text{PO}_4$	Sodium Nitrate	NaNO_3
		Potassium Nitrate	KNO_3
Calcium Nitrate	$\text{Ca}(\text{NO}_3)_2$	Potassium Sulphate	K_2SO_4
		Urea	$\text{NH}_2\text{-CO-NH}_2$

Table 7.3 Some Common Fertilizers

Fertilizers are produced by chemical reaction e.g. Ammonium nitrate is produced by the reaction of ammonia with nitric acid.

HARMFUL EFFECTS OF IMPROPER USE OF FERTILIZERS

- ✓ Discuss harmful effects of improper use of fertilizers.

Excessive fertilizers may weaken a plant, promote disease, invite pests and harming the environment. So consider plant needs carefully before applying any fertilizer.

Long term use of chemical fertilizers can be harmful to soil. Fertilizers should be used according to the amount suggested by the agricultural experts.

Excessive use of fertilizers can lead to the loss of organic material humus. The top layer of soil is fertile. It gets destroyed by excessive use of fertilizer because it becomes dry, powdery and not feasible for fertilizer. The excessive use of nitrate in the soil can raise the level of free nitrate in the fruits and crops, which is harmful to people eating them.

HYDROGENATION

- ✓ Describe the chemical process in which vegetable oil changes into fat.

DO YOU KNOW?

Humus is blackish brown coloured organic material formed by decaying of leaves, plants, and bodies of animals.

Chemical process which changes vegetable oil into fat is called hydrogenation. The vegetable oil, is unsaturated it can add more hydrogen to its carbon backbone. When Hydrogen is passed through vegetable oil in the presence of nickel, it converts into solid nickel. In this way unsaturated vegetable oil converts into saturated fats and become, solid. Hydrogenation process is used for changing oil into banaspti ghee. After the reaction the nickel is removed. This chemical process makes them suitable for making margarine, cakes and pastries etc.

Vegetable oil + Hydrogen $\xrightarrow{\text{Nickel}}$ Fat (Banaspti Ghee)

PLASTICS

- ✓ Describe the simple process for the manufacture of plastics.

DO YOU KNOW?

Poly means many; Mono means one and Mer means part.

Plastic is a synthetic material which can be easily shaped. These are polymers made up of many small identical molecules called monomers, all strung together chemically to form complex structures.

Plastics are polymers. Polymers are somewhere between a liquid and a solid, molding to the shape of their container, but stretchy and moldable like a solid. Plastics are usually tough and good electric insulators.

Following is the simplest method for making plastic.

Activity 7.3: Making Simple Plastic.

What I need:

- White glue.
- 2 bowls
- Water
- Borax powder (laundry detergent)
- Plastic spoon for stirring

What to do:

- Dissolve one tea spoon of borax in water in the bowl.
- Add half cup glue and half cup water in next bowl stir and mix it thoroughly.
- Now add the glue mixture to borax solution and stir slowly.
- Add a few drops of food coloring and mix it up.
- The slime/polymer will begin to form immediately. Stir as much as you can, then knead it with your hands until it becomes less sticky. If water is left in the bowl, just pour it out.
- Store the slime in a secure bag in a fridge, to avoid growing mold.
- The white glue has an ingredient called polyvinyl acetate, which is a liquid polymer.
- The slimy polymer plastic is fine to touch with your bare hands. Play with it, stretch it, mold it, make crazy shapes. It's difficult to get it to hold it's shape, but it makes a great creepy plastic alien to play with.

EXPLORATION

Observe your surroundings and make a list of 5 reversible and 5 irreversible changes happening around you. Discuss and share them with your teachers, class mates, elder sisters and brothers and parents.

Summary

- Changes in materials are of two types:
 1. Physical change
 2. Chemical change
- Physical change is a change in physical appearance or state of a substance.
- Chemical change is a change in chemical composition of a substance.
- Chemical fertilizers, banaspati ghee and plastics are the product of chemical changes.
- Physical and chemical properties of fertilizers are very important for the growth of plants because they overcome the deficiency of nutrients in the soil.
- Improper use of fertilizers can cause water pollution.
- Vegetable oil converts into banaspati ghee by the process of hydrogenation in the presence of nickel as catalyst.
- Plastics are very large molecules called polymers which are made from many smaller molecules called monomers.

Review Exercises

1. Write short answers of the following questions:

- i) Differentiate between physical and chemical changes by giving at least two examples of each.
- ii) Write three examples of physical changes and three examples of chemical changes which you have observed in your home/school or locality.
- iii) Identify the type of change:
 - A) Dissolving a solid (sugar) in water.
 - b) Mixing things together and
 - c) Separating mixtures.

2. Give reason(s) for your answer(s).

- i) Explain the use of hydrocarbon as fuels.
- ii) Why fertilizers are used by farmers? What happens if fertilizers are used improperly?
- iii) Describe the common properties of fertilizers.
- iv) Describe the simple process of making plastic.
- v) Explain how vegetable oil changes into fat?

3. Fill in the blanks with suitable words

- i) Melting of ice cream is a _____ change.
- ii) The _____ change is irreversible.
- iii) Mixing of sugar in water is a _____ change.
- iv) Melting of wax is a _____ change.
- v) _____ changes are reversible.

4. Choose the correct answer from the given choices.

i) What is produced when fuel is burned?

- (a) Carbon dioxide gas (b) Methane gas (c) Oxygen gas

ii) Burning of wood is a _____ change.

- (a) Chemical (b) Physical (c) Temporary

iii) Oil can change into fat by _____.

- (a) Evaporation (b) Fertilization (c) Hydrogenation

iv) Fertilizers are used to fulfil the _____ requirements of plants.

- (a) Nutritional (b) Reproductive (c) Respiratory

5. Identify and tick the reversible and irreversible change in the following:

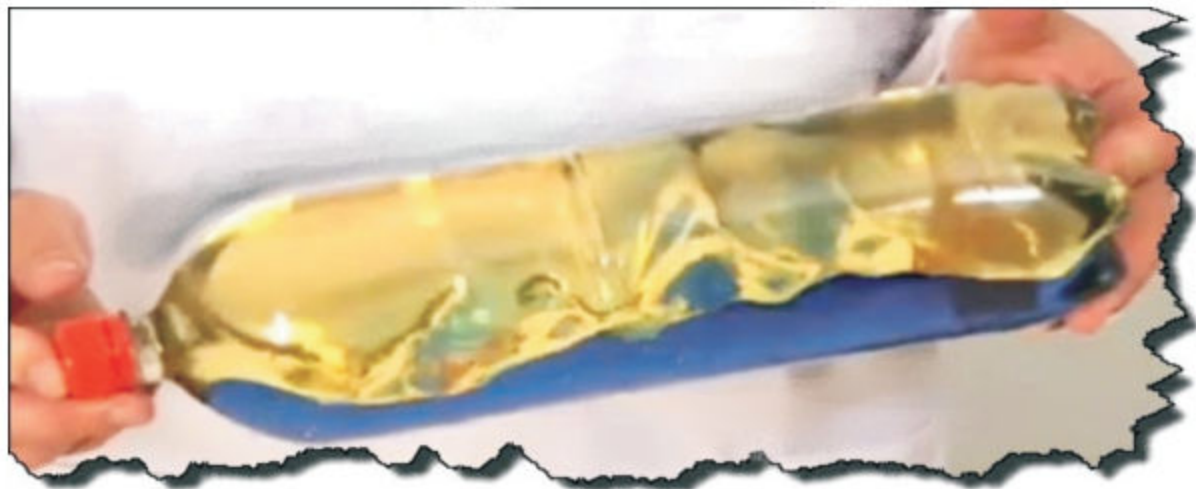
Sr. No.	Statement	Reversible Change	Irreversible Change
1.	Bending of rubber		
2.	Burning of wax		
3.	Breaking of glass		
4.	Photosynthesis		
5.	Production of glucose		
6.	Water from Hydrogen and Oxygen		
7.	Conversion of ice into water		
8.	Changing of paper into ash		
9.	Boiling of an egg		
10.	A mixture of salt and sand		
11.	Glowing of electrical bulb		
12.	Dissolving sugar into water		

PROJECT

OCEAN IN A PLASTIC BOTTLE

What I need:

- Plastic bottle 1.5 litre
- Water
- Cooking oil
- Blue food-colour



What to do:

1. Take a clear plastic bottle with cap.
2. Fill $\frac{1}{3}$ rd of bottle with water.
3. Pour few drops of blue food-colouring in water and mix it well.
4. Now fill the bottle with cooking oil and place the cap tightly.
5. Hold the bottle horizontally in both the hands and shake it.
6. Observe carefully, does the movement of water resemble the movement of water in the ocean? Yes/No, Why?
7. Share and discuss your answer with the whole class.

PROJECT

MAKE THE COLOURS DANCE

What I need:

- Fresh/Full cream milk
- A small dish/ saucer
- Four different colours of food colouring
- Liquid dish wash soap
- A dropper

What to do:

- Take some milk into a dish. (The more fat in the milk the more effective the results).
- Pour one drop of food colouring by the dropper anywhere on the milk, but not in the centre of the dish/saucer.
- Repeat this with the other three colours, make sure not to get the other colours too close to one another.
- Now carefully add one drop of dish washing liquid soap in the center of the milk and watch what happens.

What I observed:

Activity questions:

- Why full cream milk is used in this activity?
- What role does a drop of water play in this activity?
- Why colours start dancing?
- What chemical changes take place in this activity?

What I Conclude: